

5	(a)			magnetic force per unit current per unit length acting on a current-carrying long straight conductor placed at right angle to the magnetic field.	B1
	(b)			<u>charged particle (electrons)</u> moving <u>perpendicularly to magnetic field</u> experience force indicate direction towards face QRGF e.g. P to Q, E to F	B1 B1
	(c)	(i)		deflected electrons <u>build-up</u> on one face so <u>lower potential</u> than opposite face	M1 A1
		(ii)		Face <u>QRGF</u> and face <u>PSHE</u>	A1
	(d)	(i)		$\text{density} = \frac{2.7 \times 10^{-3}}{(1 \times 10^{-2})^3} = 2.7 \times 10^3 \text{ kg m}^{-3}$ $\text{number per unit volume} = \frac{2.7 \times 10^3}{4.5 \times 10^{-26}} = 6.0 \times 10^{28} \text{ m}^{-3}$	C1 M1 A0
		(ii)		$V_H = \frac{0.15 \times 4.6}{6.0 \times 10^{28} \times 0.090 \times 10^{-3} \times 1.60 \times 10^{-19}}$ $= 8.0 \times 10^{-7} \text{ V}$	M1 A1
					[Total: 10]